M-PLICITS

Neural Implicit Surfaces via Nested Multiscale Residuals

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Paper + code · Animation · NeurIPS 2026

CONTINUOUS SDF · **NESTED RESIDUALS** · **NOISE ROBUSTNESS** · **FASTER INFERENCE** · **NEURAL ATTRIBUTES**

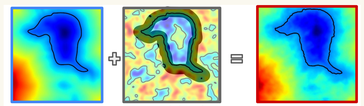
01 / REPRESENTATION

Why localize neural surfaces?

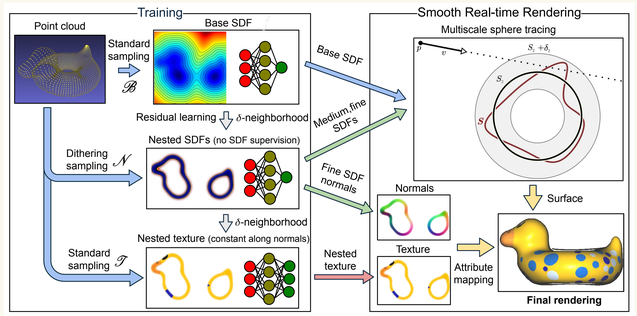
Single large MLP
Fine detail needs large models; expensive inference per query.

Grid-based encoding
Local evaluation, but can overfit input noise.

Our idea: nested residuals



One hierarchy for geometry and appearance



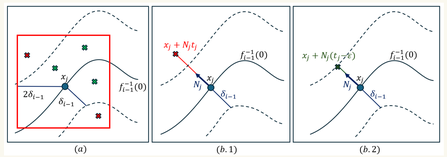
$$S_i = f_i^{-1}(0) \qquad f_{i+1} = f_i + r_i$$

$$\Omega_{i+1} = \Omega_i \cap \{x : |f_i(x)| < \delta_i\} \qquad \delta_i = (1+\varepsilon)\max_j |f_i(x_j)|$$

Surface + normal constraints; Eikonal within each band.

$$f_i(x_j) \approx 0, \quad \nabla f_i(x_j) \approx N_j, \quad \|\nabla f_i\| \approx 1$$

Local supervision, richer samples



Jitter for Eikonal; normal offsets for distance supervision.

02 / RECONSTRUCTION

Recover shape from noisy oriented points



Normal-direction perturbations up to 1% of the bounding box.
Coarse smoothness supplies the prior; residuals recover detail.

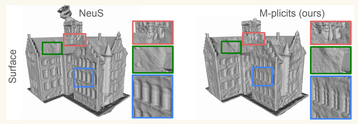
Noise robustness: paper Table 3

Stanford + Thingi32 · 1% noise · best values in bold

Method	mean CD ↓	median CD ↓	mean IoU ↑	median IoU ↑
Ours coarse	7.42E-05	5.33E-05	3.81E-01	4.03E-01
Ours medium	4.62E-05	3.03E-05	5.37E-01	**6.38E-01**
Ours fine	**3.36E-05**	**1.62E-05**	**5.82E-01**	5.91E-01
iNGP coarse	6.70E-04	3.76E-04	1.90E-01	1.88E-01
iNGP fine	2.80E-04	1.09E-04	3.09E-01	3.05E-01
IDF	8.37E-02	2.49E-05	3.22E-01	3.08E-01
BACON	1.68E-03	1.51E-04	1.76E-01	1.65E-01

CD: Chamfer distance. IoU: intersection over union.

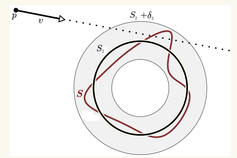
Preliminary: reconstruction from posed images



NeuS with a two-scale SDF. DTU Scan 24. 37% fewer SDF parameters.

03 / ALGORITHMS + ATTRIBUTES

Multiscale sphere tracing

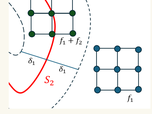


Trace coarse offsets first; use finer SDFs near the surface.

$$p_{k+1} = p_k + v(f_j(p_k) - \delta_j)$$

1.8× to 9.5× faster
Full (3 levels) to coarse · vs. SIREN · Tab. 4

Surface extraction with band culling



Evaluate the coarse grid; query fine residuals only inside the bands.

2.8× to 4.9× faster
Marching cubes vs. SIREN · Tab. 5

GEMM-based analytical normals for MLPs

$$G_i = (W_i G_{i-1}) \odot \varphi'(a_i)$$

Batched chain-rule gradients; no autograd graph.
≈2× faster than PyTorch autograd (paper supplement).

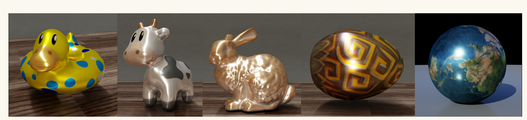
Normal mapping vs. silhouette refinement

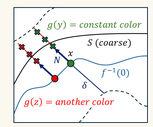


Coarse → + normals · + tracing → silhouette

Fine normals add detail. Extra tracing refines the silhouette.

Texture fields without UV parameterization





Fit RGB on points; extend color along normals.

$$\nabla g \cdot \nabla f \approx 0$$

246,627 parameters · base + 2 residual SIRENs

Paper, code, data and fitted models

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